3. Functional Module Descriptions

3.1 Authentication and Session Management

Authentication is the entry point for all agent journeys. Any defect here blocks access to the platform and has high business impact.

Attribute	Detail
Primary pages	/login, /forgot-password, /dashboard
User types	Valid agent, new agent, locked agent, MFA agent, admin user, invalid user
Primary risks	Login failure, weak validation, incorrect error messages, unauthorized access, account lockout not enforced, session not cleared after logout
Automation focus	Positive and negative login, form validation, logout, session timeout, protected route access, lockout, MFA/OTP simulation

Validation ID	Scenario	Automation Approach
AUTH-V01	Valid agent can log in successfully.	Use LoginPage object; assert dashboard URL and user greeting.
AUTH-V02	Invalid password displays correct error message.	Submit wrong password; assert visible error toast or inline message.
AUTH-V03	Blank username and password show mandatory validation.	Submit empty form; assert field-level errors.
AUTH-V04	Logout clears session and redirects to login page.	Login, logout, then assert protected route redirects.
AUTH-V05	Direct access to dashboard without login is blocked.	Open /dashboard in a fresh context and verify redirect.
AUTH-V06	Locked user cannot access dashboard.	Login with locked user; assert account locked message.
AUTH-V07	Account locks after five failed login attempts.	Attempt invalid login five times; assert locked state.
AUTH-V08	Password policy rejects weak password.	Enter weak password during reset/change password and assert validation.

3.2 Agent Dashboard and Application Summary

The dashboard gives agents a consolidated view of draft applications, submitted applications, pending actions, and status tracking.

Validation ID	Scenario	Automation Approach
DASH-V01	Dashboard loads after successful login.	Assert draft summary, submitted summary, and new application CTA are visible.
DASH-V02	Agent sees only own applications.	Login as different agents and validate data isolation.
DASH-V03	New agent sees empty application state.	Login as new agent; assert empty state copy and CTA.
DASH-V04	Draft application can be resumed.	Open draft from dashboard and assert previously entered data.
DASH-V05	Application status filter works correctly.	Apply Draft/Submitted/Approved filters and validate rows.
DASH-V06	Dashboard application status matches API response.	Compare displayed status with application status API.

3.3 Client / Insured Information

Client information capture is a high-risk module because incorrect client data can affect eligibility, compliance, and policy issuance.

Feature	Expected Behaviour	QA Focus
Personal details	Mandatory name, DOB, gender, marital status, nationality, occupation, and employer fields are captured.	Mandatory fields, format validation, DOB boundaries.
Age calculation	Age is calculated from date of birth.	Age between 18 and 65, future DOB rejection.
Address	Residential and mailing address are captured.	Required fields, postal code format, same-as-residential behavior.
Contact information	Email, mobile, alternate phone, and preferred communication mode are captured.	Email/phone format and mandatory contact method.
KYC	ID type, ID number, issue/expiry date, and document upload are captured.	Dynamic ID validation, SSN masking, upload validation.
Financial suitability	Income, net worth, source of funds, and risk profile are captured.	Income versus premium affordability validation.
Compliance	PEP declaration and related details are captured.	Incomplete declaration blocking and PEP flagging.

Validation ID	Scenario	Automation Approach
CLI-V01	Agent can enter valid insured personal information.	Fill valid data; assert next step is enabled.
CLI-V02	Mandatory personal fields are validated.	Leave required fields blank; assert inline errors.
CLI-V03	DOB below 18 years is rejected.	Enter underage DOB and assert eligibility error.
CLI-V04	DOB above 65 years is rejected.	Enter age above eligibility and assert error.
CLI-V05	Future DOB is rejected.	Enter future DOB and assert validation.
CLI-V06	Same-as-residential address copies data to mailing address.	Fill residential address, select checkbox, assert copied fields.
CLI-V07	Invalid email and phone formats are rejected.	Enter invalid formats and assert errors.
CLI-V08	SSN is masked after entry.	Enter SSN and assert only last four digits are visible.
CLI-V09	KYC upload rejects unsupported file type.	Upload invalid file and assert error.
CLI-V10	PEP Yes requires additional details.	Select PEP Yes and assert details field is mandatory.

3.4 Product Selection and Policy Configuration

Product selection allows agents to select Whole Life products and configure policy details.

Feature	Expected Behaviour	QA Focus
Product catalog	Whole Life products are listed with key details.	Product visibility, active products, detail accuracy.
Product details	Features, eligibility, coverage limits, and benefits are visible.	Field validation and navigation.
Face value	Sum assured can be configured within product limits.	Minimum, maximum, and boundary validation.
Premium frequency	Monthly, quarterly, semi-annual, and annual options are supported.	Premium recalculation and review page consistency.
Payment term	Payment term can be selected as per product rules.	Dropdown values and rule validation.
Policy term	Whole Life / to Age 100 type rules are supported.	Configuration and display validation.
Premium calculation	Premium updates in real time.	UI calculation, itemization, and API reconciliation.

Validation ID	Scenario	Automation Approach
PROD-V01	Product catalog displays expected Whole Life products.	Assert product cards and product names.
PROD-V02	Product details page opens correctly.	Click product and assert features, eligibility, and coverage limits.
PROD-V03	Face value below minimum is rejected.	Enter below-minimum amount and assert validation.
PROD-V04	Face value above maximum is rejected.	Enter above-maximum amount and assert validation.
PROD-V05	Premium recalculates when frequency changes.	Change frequency and assert premium value changes.
PROD-V06	Premium displayed on review page matches configuration page.	Capture premium and compare on review page.
PROD-V07	Product suitability warning appears for mismatch.	Use unsuitable risk profile and assert warning.

3.5 Rider Selection

Rider selection tests important business rules involving eligibility, dependencies, maximum coverage, and mutual exclusivity.

Validation ID	Scenario	Automation Approach
RIDER-V01	Agent can add an eligible protection rider.	Select ADB rider and assert rider appears in premium summary.
RIDER-V02	Rider coverage cannot exceed base policy face value.	Enter excessive rider coverage and assert validation.
RIDER-V03	Age-ineligible rider cannot be added.	Use client age outside allowed range and assert disabled or error state.
RIDER-V04	Maximum rider count is enforced.	Attempt to add more than allowed rider count and assert blocking message.
RIDER-V05	Mutually exclusive waiver riders cannot both be selected.	Select one waiver rider and assert other is disabled or blocked.
RIDER-V06	Critical illness rider triggers health questionnaire.	Select rider and assert questionnaire section appears.
RIDER-V07	Rider premium is itemized in total premium.	Add rider and assert itemized premium and total premium update.

3.6 Beneficiary Management

Beneficiary management is a strong automation area because it includes dynamic forms, allocation validation, minor rules, beneficiary types, and duplicate prevention.

Feature	Expected Behaviour	QA Focus
Primary beneficiary	At least one primary beneficiary is mandatory.	Submission blocking and validation summary.
Allocation percentage	Primary and secondary allocation totals must equal 100%.	Under-allocation, over-allocation, running total.
Multiple beneficiaries	Add, edit, and remove beneficiaries dynamically.	Dynamic row/card handling and persistence.
Beneficiary type	Individual, Trust, and Organization types show relevant fields.	Conditional fields and mandatory validations.
Minor beneficiary	Guardian details are mandatory when beneficiary is below 18.	DOB calculation and guardian validation.
Special designation	Irrevocable beneficiary triggers legal confirmation.	Confirmation modal and acknowledgment.
Duplicate prevention	Duplicate based on Name + DOB + Relationship is blocked.	Negative duplicate test.

Validation ID	Scenario	Automation Approach
BEN-V01	Add a valid primary beneficiary.	Fill beneficiary form and assert beneficiary card/row appears.
BEN-V02	Primary allocation below 100% blocks progression.	Enter allocation total below 100 and assert error.
BEN-V03	Primary allocation above 100% blocks progression.	Enter allocation total above 100 and assert error.
BEN-V04	Secondary allocation is validated separately.	Add secondary beneficiaries and assert separate total validation.
BEN-V05	Minor beneficiary requires guardian details.	Enter minor DOB and assert guardian fields mandatory.
BEN-V06	Trust beneficiary shows trust-specific fields.	Select Trust and assert Trust Name, Trustee Name, Trust Date.
BEN-V07	Organization beneficiary shows organization-specific fields.	Select Organization and assert registration and contact fields.
BEN-V08	Duplicate beneficiary is blocked.	Add duplicate beneficiary and assert duplicate error.
BEN-V09	Removing beneficiary updates allocation total.	Remove one beneficiary and assert running total changes.

3.7 Document Upload and Validation Summary

Document upload and validation summary ensure the application is complete before submission.

Validation ID	Scenario	Automation Approach
DOC-V01	Valid KYC document can be uploaded.	Upload PDF/JPG/PNG and assert uploaded status.
DOC-V02	Unsupported file type is rejected.	Upload invalid file and assert error.
DOC-V03	File above maximum size is rejected.	Upload oversized file and assert size validation.
DOC-V04	Missing mandatory document blocks submission.	Skip upload and assert validation summary error.
VAL-V01	Validation summary lists all unresolved errors.	Submit incomplete application and assert grouped errors.
VAL-V02	Clicking validation error navigates to corresponding field.	Click validation summary item and assert field focus/section.
VAL-V03	Submit button remains disabled until critical errors are resolved.	Leave critical errors and assert button disabled.

3.8 Application Submission and Status Tracking

Submission is the core E2E flow and should be part of the minimum regression suite.

Sample flow: Login -> Dashboard -> Start New Application -> Enter Client Information -> Select Product -> Configure Policy -> Add Riders -> Add Beneficiaries -> Upload Documents -> Review Validation Summary -> Submit -> Verify Reference Number -> Check Dashboard Status.

Validation ID	Scenario	Automation Approach
WF-V01	Agent can save incomplete application as draft.	Enter partial data, save draft, assert dashboard draft row.
WF-V02	Draft application can be resumed with saved data.	Resume draft and assert previously entered data.
WF-V03	Valid application can be submitted successfully.	Execute full E2E flow and assert reference number.
WF-V04	Submitted application becomes read-only.	Open submitted application and assert fields are not editable.
WF-V05	Confirmation email simulation/status message appears after submission.	Submit and assert confirmation message.
WF-V06	Dashboard status updates after submission.	Submit application and assert dashboard status is Submitted.
WF-V07	Application detail status matches API response.	Call API and compare application detail page status.

3.9 API Validation Scope

API tests should complement UI tests. Learners should validate API contracts and use API responses to cross-check UI outputs.

API Area	Example Endpoint	Validation Focus
Authentication	POST /api/auth/login	Status codes, token/session response, invalid credentials.
Agent Dashboard	GET /api/applications	Draft/submitted applications, status, data isolation.
Client Information	GET/POST /api/applications/{id}/client	Required fields, validation errors, persistence.
Products	GET /api/products	Product listing, eligibility, schema, required fields.
Product Details	GET /api/products/{productId}	Coverage limits, terms, eligibility, feature details.
Riders	GET /api/riders	Rider categories, eligibility, dependency rules.
Application Validation	POST /api/applications/{id}/validate	Mandatory fields, cross-field rules, warning/error structure.
Beneficiaries	GET/POST /api/applications/{id}/beneficiaries	Allocation, minor guardian, duplicate prevention.
Documents	POST /api/applications/{id}/documents	File type, file size, required document status.
Submission	POST /api/applications/{id}/submit	Reference number, status transition, validation blocking.
Status	GET /api/applications/{id}/status	Current status, status history, timestamps.

Validation ID	Scenario	Automation Approach
API-V01	Products API returns 200 and expected schema.	Use Playwright request context and assert response keys.
API-V02	Application API requires authorization.	Call without token/session and assert 401 or 403.
API-V03	Client API rejects invalid DOB.	Post future/underage DOB and assert validation response.
API-V04	Validation API returns beneficiary allocation error.	Submit under-allocation and assert validation response.
API-V05	Submission API rejects incomplete application.	Post submit request for incomplete draft and assert error.
API-V06	Application status API matches dashboard.	Compare API status with UI dashboard status.

API-V07	Product API supports product detail retrieval.	Fetch selected product and compare with UI detail page.
API-V08	Applications API returns only logged-in agent records.	Validate data isolation with multiple agents.